**Stochastic Processes — 1060429**

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Throughout, a Brownian motion (BM) is standard ($B_0 = 0$) with continuous paths, and $\mathcal{F}_t$ denotes the filtration in use. Apparent typos are pointed out where they occur.

# Exercise 1

## Relations between stochastic processes

**1.** Let $\Omega_0 = \{X, Y \text{ right-continuous}\} \cap \bigcap_{q\in\mathbb{Q}_+}\{X_q = Y_q\}$. The first event contains a measurable set of probability 1, and each $\{X_q = Y_q\}$ has probability 1 (modification), so $\Omega_0$ contains a measurable set of probability 1 (countable intersection). For $\omega \in \Omega_0$ and $t \geq 0$ take $q_n \in \mathbb{Q}_+$, $q_n \downarrow t$; then $X_t(\omega) = \lim X_{q_n}(\omega) = \lim Y_{q_n}(\omega) = Y_t(\omega)$. Hence $X_t = Y_t$ for all $t$ on $\Omega_0$, i.e. $X, Y$ are indistinguishable.

**2.** Let $Q_0 = \mathbb{Q} \cap [0, t_0)$ and

$$C_{t_0} = \bigcap_{k\geq 1}\bigcup_{m\geq 1}\bigcap_{s,u\in Q_0,\ |s-u|<1/m} \{|X_s - X_u| \leq 1/k\} \in \sigma(X_t : t \leq t_0).$$

*Claim.* If the path $X(\omega)$ is RCLL on $[0, t_0]$, then $\omega \in A_{t_0} \iff \omega \in C_{t_0}$.

If $X(\omega)$ is continuous on $[0, t_0)$ and has a left limit at $t_0$, it extends continuously to $[0, t_0]$, so it is uniformly continuous on $[0, t_0)$ and $\omega \in C_{t_0}$. Conversely, let $\omega \in C_{t_0}$, $k \geq 1$, and $m$ as in the definition. For $s, u \in [0, t_0)$ with $|s - u| < 1/m$ choose $q_j, r_j \in Q_0$ with $q_j \downarrow s$, $r_j \downarrow u$ and $|q_j - r_j| < 1/m$ (possible as $s, u < t_0$). By right-continuity $|X_s - X_u| = \lim |X_{q_j} - X_{r_j}| \leq 1/k$. So $X(\omega)$ is uniformly continuous on $[0, t_0)$, hence $\omega \in A_{t_0}$.

**(a)** If every path is RCLL, the claim gives $A_{t_0} = C_{t_0} \in \sigma(X_t : t \leq t_0)$.

**(b)** Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, $\mathbb{P} = \text{Leb}$, and let $C$ be the Cantor set. Since $|\mathcal{P}(C)| = 2^{\mathfrak{c}} > \mathfrak{c} = |\mathcal{B}([0, 1])|$, there is $E \subset C$ with $E \notin \mathcal{F}$. Define

$$X_t(\omega) = \mathbf{1}_E(\omega)\mathbf{1}_{\{t=\omega\}} + \mathbf{1}_{C\setminus E}(\omega)\mathbf{1}_{\{t=\omega+3\}}.$$

Then $X_t = \mathbf{1}_{E\cap\{t\}} + \mathbf{1}_{(C\setminus E)\cap\{t-3\}}$ is a sum of indicators of sets with at most one point, so it is measurable. For $\omega \notin C$ the path is $\equiv 0$. For $\omega \in E$ the path has a spike at $t = \omega \in [0, 1]$. For $\omega \in C \setminus E$ it has a spike at $t = \omega + 3 \geq 3$. A spike is a point where right-continuity fails. Hence $\{X \text{ RCLL}\} = \Omega \setminus C \in \mathcal{F}$ has probability 1, so $X$ is RCLL with probability 1 under any reading of this phrase. On the other hand, a path is continuous on $[0, 2)$ iff $\omega \notin E$, so $A_2 = \Omega \setminus E \notin \mathcal{F}$. In particular $A_2 \notin \sigma(X_t : t \leq 2) \subseteq \mathcal{F}$.

**(c)** There is $N \in \mathcal{F}$ with $\mathbb{P}(N) = 0$ such that every path outside $N$ is RCLL. By the claim, $A_{t_0} \triangle C_{t_0} \subseteq N$. Completeness of $\mathcal{F}_{t_0}$ (as usual, this means $\mathcal{F}_{t_0}$ contains every subset of every $\mathbb{P}$-null set $N \in \mathcal{F}$) gives $A_{t_0} \setminus C_{t_0}$, $C_{t_0} \setminus A_{t_0} \in \mathcal{F}_{t_0}$. Hence $A_{t_0} = (C_{t_0} \setminus (C_{t_0} \setminus A_{t_0})) \cup (A_{t_0} \setminus C_{t_0}) \in \mathcal{F}_{t_0}$.

## Brownian motion

**1.**

*Remark.* The condition should read $\mathbb{E}[|X_t - X_s|^\alpha] \leq M|t - s|^{1+\beta}$ for all $t, s \in [0, T]$. In (c) the bound should be $2^{-\gamma n}$, not $2^{-m}$. In (e),(f) the index set is $[0, 1]$.

**(a)** Put $\tilde{X}_u = X_{uT}$, $u \in [0, 1]$. Then $\mathbb{E}|\tilde{X}_u - \tilde{X}_v|^\alpha \leq MT^{1+\beta}|u - v|^{1+\beta}$, which is the hypothesis on $[0, 1]$ with constant $MT^{1+\beta}$. If $\tilde{Y}$ is a modification of $\tilde{X}$ that is a.s. $\gamma$-Hölder, then $Y_t = \tilde{Y}_{t/T}$ satisfies $\mathbb{P}(Y_t = X_t) = \mathbb{P}(\tilde{Y}_{t/T} = \tilde{X}_{t/T}) = 1$, and $|Y_t - Y_s| \leq K|t/T - s/T|^\gamma = KT^{-\gamma}|t - s|^\gamma$.

**(b)** By Markov's inequality, $\mathbb{P}(|X_t - X_s| > \varepsilon) \le \varepsilon^{-\alpha} M |t-s|^{1+\beta} \to 0$ as $t \to s$.

**(c)** By a union bound and Markov's inequality,

$$\mathbb{P}\Big( \max_{1 \le k \le 2^n} |X_{k2^{-n}} - X_{(k-1)2^{-n}}| > 2^{-\gamma n} \Big) \le 2^n \cdot 2^{\gamma \alpha n} M 2^{-n(1+\beta)} = M 2^{-n(\beta - \alpha\gamma)},$$

which is summable since $\gamma < \beta/\alpha$. By Borel–Cantelli, a.s. only finitely many of these events $E_n$ occur. Let $N(\omega)$ be the last $n$ at which one occurs ($N = 0$ if none occurs). $N$ is measurable, since $\{N \le k\} = \bigcap_{n>k} E_n^c$. Then $\mathbb{P}(N < \infty) = 1$ and $\max_k |X_{k2^{-n}} - X_{(k-1)2^{-n}}| \le 2^{-\gamma n}$ for all $n > N(\omega)$.

**(d)** Fix $\omega$ with $N(\omega) < \infty$. *Claim:* for $m > n \ge N(\omega)$ and $s < t$ in $D_m$ with $t - s < 2^{-n}$,

$$|X_t - X_s| \le 2 \sum_{j=n+1}^{m} 2^{-\gamma j}.$$

We use induction on $m$. For $m = n+1$ we have $t - s = 2^{-(n+1)}$, and the bound holds by (c). For the step $m \to m+1$, let $s, t \in D_{m+1}$. Let $s' = \min\{r \in D_m : r \ge s\}$ and $t' = \max\{r \in D_m : r \le t\}$. Then $s' - s,\ t - t' \in \{0, 2^{-(m+1)}\}$, and $s' \le t'$ (otherwise $s < t < s' \le s + 2^{-(m+1)}$, which is impossible in $D_{m+1}$). Since $t' - s' \le t - s < 2^{-n}$, the induction hypothesis (or $t' = s'$) and (c) at level $m+1 > N$ give

$$|X_t - X_s| \le 2 \sum_{j=n+1}^{m} 2^{-\gamma j} + 2 \cdot 2^{-\gamma(m+1)}.$$

Now let $s < t$ in $D = \bigcup_m D_m$ with $t - s < 2^{-N(\omega)}$. Choose $n$ with $2^{-(n+1)} \le t - s < 2^{-n}$, so $n \ge N(\omega)$, and choose $m > n$ with $s, t \in D_m$. Then

$$|X_t - X_s| \le 2 \sum_{j>n} 2^{-\gamma j} = \frac{2 \cdot 2^{-\gamma(n+1)}}{1 - 2^{-\gamma}} \le C_\gamma |t-s|^\gamma, \qquad C_\gamma = \frac{2}{1 - 2^{-\gamma}}.$$

**(e)** Let $s < t$ in $D$ with $t - s \ge 2^{-N}$. The points of $D_{N+1} \cap (s,t)$ together with $s, t$ form a chain $s = u_0 < \cdots < u_k = t$ in $D$ with consecutive gaps $< 2^{-N}$ and $k \le 2^{N+1} + 1$. Each gap contributes at most $C_\gamma 2^{-N\gamma} \le C_\gamma |t-s|^\gamma$. Hence on $\{N < \infty\}$,

$$|X_t - X_s| \le K(\omega) |t-s|^\gamma \ \text{ for all } s, t \in D, \qquad K = (2^{N+1} + 1) C_\gamma.$$

**(f)** Let

$$\Omega' = \bigcap_{k \ge 1} \bigcup_{m \ge 1} \bigcap_{r, r' \in D,\ |r - r'| < 1/m} \{|X_r - X_{r'}| \le 1/k\} \in \mathcal{F}$$

be the event that $X|_D$ is uniformly continuous. By (e), $\Omega' \supseteq \Omega^*_\gamma := \{N_\gamma < \infty\}$ for every $\gamma < \beta/\alpha$, where $N_\gamma$ is the variable $N$ of (c) for this $\gamma$. On $\Omega'$ the limit $Y_t := \lim_{D \ni r \to t} X_r$ exists for all $t \in [0,1]$, since uniformly continuous functions on a dense set extend continuously. Set $Y \equiv 0$ off $\Omega'$. Each $Y_t$ is measurable (a limit along a sequence), $Y$ has continuous paths, and $Y$ does not depend on $\gamma$. On $\Omega^*_\gamma$, $Y$ is $\gamma$-Hölder on $[0,1]$ with constant $K$, by (e) and continuity. For fixed $t$ take $r_j \in D$, $r_j \to t$: then $X_{r_j} \to Y_t$ a.s. and $X_{r_j} \to X_t$ in probability by (b). Limits in probability are a.s. unique, so $Y_t = X_t$ a.s., and $Y$ is a modification of $X$. Take $\gamma_k \uparrow \beta/\alpha$. On the probability-one event $\bigcap_k \Omega^*_{\gamma_k}$, $Y$ is $\gamma_k$-Hölder for every $k$. For $\gamma < \gamma_k$ and $|t-s| \le 1$ we have $|t-s|^{\gamma_k} \le |t-s|^\gamma$, so $\gamma_k$-Hölder on $[0,1]$ implies $\gamma$-Hölder. Hence a.s. $Y$ is $\gamma$-Hölder for every $\gamma < \beta/\alpha$.

**2.** Here $t_0$ is a local maximum if there is $\varepsilon > 0$ with $B_s \le B_{t_0}$ for all $s \in [0,\infty) \cap (t_0 - \varepsilon, t_0 + \varepsilon)$.

**(a)** $W_s = B_{t_0+s} - B_{t_0}$ is a BM. If $t_0$ is a local maximum, then $\max_{[0,1/n]} W \le 0$ for some $n$. By the reflection principle,

$$\mathbb{P}\big( \max_{[0,1/n]} W \ge a \big) = 2\mathbb{P}(W_{1/n} \ge a) \xrightarrow[a \downarrow 0]{} 1,$$

so $\mathbb{P}(\max_{[0,1/n]} W > 0) = 1$ for every $n$. Hence $\mathbb{P}\big(\bigcup_n \{\max_{[0,1/n]} W \le 0\}\big) = 0$, and $\{t_0 \text{ local max}\}$ is contained in a null set.

**(b)** Let $M = \max_{[0,1]} B$ and $G = \{M > B_0\} \cap \{M > B_1\} \in \mathcal{F}$. If $M = B_0$ then 0 is a local maximum, so $\mathbb{P}(M \le B_0) = 0$ by (a) with $t_0 = 0$. The process $W_s = B_1 - B_{1-s}$, $s \in [0,1]$, is a continuous centred Gaussian process. Its covariance is

$$\operatorname{Cov}(W_s, W_u) = 1 - (1-u) - (1-s) + (1 - s \vee u) = s \wedge u,$$

so $W$ is a BM on $[0,1]$. Moreover $M = B_1$ means $\min_{[0,1]} W \ge 0$, i.e. $\max_{[0,1]}(-W) \le 0$, which has probability 0 by the argument of (a) applied to the BM $-W$. Hence $\mathbb{P}(G) = 1$. On $G$ every maximiser $t^*$ of $B$ on $[0,1]$ lies in $(0,1)$, so $B_s \le B_{t^*}$ for all $s$ in a neighbourhood of $t^*$, i.e. $t^*$ is a local maximum.

**3.**

**(a)** Let $Z_n = (B_{1/n} - B_{1/(n+1)})/\sqrt{1/n - 1/(n+1)}$. The intervals $[\frac{1}{n+1}, \frac{1}{n}]$ are non-overlapping (they meet only at endpoints), so by independence of increments the $Z_n$ are i.i.d. $N(0,1)$. Then $\sum_n \mathbb{P}(Z_n \ge K) = \sum_n \mathbb{P}(Z_1 \ge K) = \infty$, and the second Borel–Cantelli lemma gives $\mathbb{P}(Z_n \ge K \text{ i.o.}) = 1$.

**(b)** Fix rationals $p < q$. The process $W_t = B_{p+t} - B_p$ is a BM. Applying (a) to $W$ for every $K \in \mathbb{N}$, a.s. for all $K$ there are infinitely many $n$ with $1/n \le q - p$ and

$$|W_{1/n} - W_{1/(n+1)}| \ge K |1/n - 1/(n+1)|^{1/2}.$$

So $B$ is not $\frac{1}{2}$-Hölder on $[p,q]$. Intersecting over the countably many pairs $(p,q)$, and using that every non-trivial interval contains some $[p,q]$, gives the claim.

**4.** Let $n \in \mathbb{N}$, $h = 1/n$, $t_k = kh$, $\delta = \varepsilon/4$. Let $L$ be the piecewise linear interpolation of $(t_k, B_{t_k})_k$ and $Z = B - L$. On $[t_{k-1}, t_k]$,

$$Z_s = (B_s - B_{t_{k-1}}) - \tfrac{s - t_{k-1}}{h}(B_{t_k} - B_{t_{k-1}}), \qquad \text{so} \qquad \sup_{[t_{k-1},t_k]} |Z| \le 2 \sup_{u \in [0,h]} |B_{t_{k-1}+u} - B_{t_{k-1}}|.$$

*Independence.* $(Z_s)_s$ and $(B_{t_j})_j$ are jointly Gaussian and $\operatorname{Cov}(Z_s, B_{t_j}) = 0$. For $t_j \le t_{k-1}$ this covariance equals $t_j - \frac{(t_k - s)t_j + (s - t_{k-1})t_j}{h} = 0$, and for $t_j \ge t_k$ it equals $s - \frac{(t_k - s)t_{k-1} + (s - t_{k-1})t_k}{h} = 0$. Hence $\sigma(Z_s : s \in [0,1])$ is independent of $(B_{t_1}, \dots, B_{t_n})$.

Let $f_h$ be the linear interpolation of $(t_k, f(t_k))$. Then $|f_h - f| \le \omega_f(h)$, where $\omega_f$ is the modulus of continuity. Consider the events

$$A = \{\max_k |B_{t_k} - f(t_k)| < \delta\}, \qquad C = \{\sup_{[0,1]} |Z| < \delta\}.$$

On $A \cap C$ we have $|B_t - f(t)| \le |L_t - f_h(t)| + |f_h(t) - f(t)| + |Z_t| < 2\delta + \omega_f(h)$, which is $< \varepsilon$ once $\omega_f(h) < \varepsilon/2$.

$\mathbb{P}(A) > 0$, since $(B_{t_1}, \dots, B_{t_n})$ is an invertible linear image of i.i.d. $N(0,h)$ increments and so has an everywhere positive density. The suprema over the different intervals are independent (independent increments) and each has the law of $\sup_{[0,h]} |B|$. By scaling,

$$\mathbb{P}(C) \ge \mathbb{P}\big(\sup_{[0,h]} |B| < \delta/2\big)^n = \mathbb{P}\big(\sup_{[0,1]} |B| < \delta/(2\sqrt{h})\big)^n.$$

The inner probability tends to 1 as $h \to 0$, so it is positive for $h$ small. Choosing such $h$ with also $\omega_f(h) < \varepsilon/2$, independence gives $\mathbb{P}(\sup_{[0,1]} |B - f| < \varepsilon) \ge \mathbb{P}(A)\mathbb{P}(C) > 0$.

**5.**

**(a)** $X_0 = 0$ and $X$ has continuous paths. For $0 \le t_0 < \cdots < t_k$, the increments $X_{t_i} - X_{t_{i-1}} = a \cdot (B_{t_i} - B_{t_{i-1}})$ are functions of the independent vectors $B_{t_i} - B_{t_{i-1}}$, hence independent. Each is a sum of independent $N(0, a_j^2(t_i - t_{i-1}))$ variables, i.e. $N(0, |a|^2(t_i - t_{i-1})) = N(0, t_i - t_{i-1})$. So $X$ is a BM.

**(b)** By (a) each coordinate $Y^i = a_i \cdot B$ is a BM, where $a_i$ is the $i$-th row. All $Y^i_t$ are linear combinations of the Gaussian family $(B^j_t)$, so $(Y^1, \dots, Y^m)$ is jointly Gaussian with $\mathrm{Cov}(Y^i_s, Y^j_t) = (a_i \cdot a_j)(s \wedge t) = \delta_{ij}(s \wedge t)$. Uncorrelated jointly Gaussian families are independent: the finite-dimensional distributions factor, and a $\pi$–$\lambda$ argument finishes the proof. So $Y^1, \dots, Y^m$ are independent BMs.

**(c)** For $R \in SO(d)$ the rows are orthonormal, so by (b) $RB$ consists of $d$ independent BMs, i.e. it is a $d$-dimensional BM. Since the law of a process is determined by its finite-dimensional distributions, $(RB_t)_t \overset{d}{=} (B_t)_t$.

**6.** For a partition $\pi$ of $[0, T]$ write $Q_\pi(X) = \sum_i (X_{t_i} - X_{t_{i-1}})^2$. We use the known result that $Q_\pi(B) \to v - u$ in $L^2$ for partitions $\pi$ of any interval $[u, v]$; indeed $\mathbb{E}(Q_\pi(B) - (v-u))^2 = 2\sum \Delta t_i^2 \le 2(v-u)|\pi|$.

**(a)** $Q_\pi(aB) = a^2 Q_\pi(B) \to a^2 T$.

**(b)** $W_t = c^{-1/2} B_{ct}$ is a BM, so $Q_\pi(B_{c\cdot}) = c Q_\pi(W) \to cT$.

**(c)** Let $Y_t = \int_0^t B_s\, ds$ and $M = \sup_{[0,T]} |B|$. Then $|Y_{t_i} - Y_{t_{i-1}}| \le M \Delta t_i$, so $Q_\pi(Y) \le M^2 T |\pi|$ and $\mathbb{E} Q_\pi(Y)^2 \le \mathbb{E}[M^4] T^2 |\pi|^2 \to 0$. Here $\mathbb{E} M^4 < \infty$, since $M \le \sup B + \sup(-B)$ and $\sup_{[0,T]} B \overset{d}{=} |B_T|$ by the reflection principle. The quadratic variation is 0.

**(d)** If $t_0 \in \pi$, then $Q_\pi(X) = Q_{\pi \cap [0,t_0]}(B) + 9\, Q_{\pi \cap [t_0, T]}(B) \to t_0 + 9(T - t_0)$ in $L^2$. Otherwise let $[t_{j-1}, t_j] \ni t_0$, $u = B_{t_0} - B_{t_{j-1}}$, $v = -3(B_{t_j} - B_{t_0})$. Then $Q_\pi(X) = Q_{\pi \cup \{t_0\}}(X) + 2uv$, and by independence $\mathbb{E}(2uv)^2 = 36(t_0 - t_{j-1})(t_j - t_0) \le 36|\pi|^2 \to 0$. Hence the quadratic variation is $t_0 + 9(T - t_0) = 9T - 8t_0$.